🔷 Appendix A — Mathematical Proofs for SEKHEM-TSDD (Final — Clean Version)

A.1 Existence and Uniqueness of Dynamics

Consider:

$$\Psi_{k+1} = \Psi_k + dt\,F(\Psi_k) + \sigma\sqrt{dt}\,\xi_k$$

Assumptions:

- F is globally Lipschitz
- $\|F(\Psi)\| \le C$
- $\Omega \subset \mathbb{R}^d$ is compact

Proof

From Lipschitz continuity:

$$\|F(x)-F(y)\| \le L\|x-y\|$$

Thus, the continuous-time system admits a unique solution by the Picard–Lindelöf theorem.

For the discrete-time Euler scheme, if:

$$dt < \frac{1}{L}$$

then the approximation is stable and convergent.

$$
\boxed{
\text{A unique solution exists and is well-defined.}
}
$$

A.2 Boundedness of the State

$$
F =
-\nabla \mathcal{E}
- L\Psi
+ \alpha(PAC)\nabla \ln P
+ B(u)
$$

Bounds:

$$
\|\nabla \mathcal{E}\| \le C_1,\quad
\|L\Psi\| \le C_2,\quad
\|\nabla \ln P\| \le C_3,\quad
\|B(u)\| \le C_4
$$

Thus:

$$
\|F\| \le C
$$

Using recursion:

$$
\|\Psi_{k+1}\|
\le
\|\Psi_k\| + dt\,C + \sigma\sqrt{dt}\|\xi_k\|
$$

Applying discrete Grönwall inequality:

$$
\|\Psi_k\| \le C'
$$

$$
\boxed{
\text{The system state remains bounded.}
}
$$

A.3 Lyapunov Stability

$$
\mathcal{L}(\Psi,P) =
\mathcal{E}(\Psi) + KL(P\|P_0)
$$

Differentiating:

$$
\dot{\mathcal{L}} =
\langle \nabla \mathcal{E}, \dot{\Psi} \rangle +
$$

$$\dot{KL}$$

Substituting system dynamics:

$$
\dot{\mathcal{L}}
\le
- c_1\|\nabla \mathcal{E}\|^2
- c_2\|L\Psi\|^2
+ c_3\|\nabla \ln P\|^2
+ c_4\|u\|^2
+ C\sigma^2
$$

Using boundedness:

$$\dot{\mathcal{L}} \le -c\|\Psi\|^2 + C\sigma^2$$

$$
\boxed{
\text{The system is globally Lyapunov stable.}
}
$$

A.4 Mean-Square Stability

$$\Psi_{k+1} = \Psi_k + dt\,F(\Psi_k) + \text{noise}$$

Taking expectation:

$$\mathbb{E}\|\Psi_{k+1}\|^2$$

$$
\le
(1 - c\,dt)\mathbb{E}\|\Psi_k\|^2 + C\sigma^2 dt
$$

Solving recursion:

$$
\mathbb{E}\|\Psi_k\|^2
\le
(1 - c\,dt)^k \|\Psi_0\|^2 + O(\sigma^2)
$$

$$
\boxed{
\text{The system is mean-square bounded and stable.}
}
$$

A.5 PAC Robustness

$$
D = \mathrm{median}(R,C,P)
$$

Consider $X=(x_1,x_2,x_3)$ with one corrupted element:
$x_3 \to \infty$

Then:

$$
D = \mathrm{median}(x_1,x_2,x_3)
$$

remains bounded.

$$\boxed{
\text{The breakdown point of the PAC fusion is }
33\%.
}$$

A.6 Decision Optimality

$$u_k = \arg\max_{u \in \mathcal{U}} D_u(k)$$

Since:

- $\mathcal{U}$ is finite
- D_u is bounded

$$\Rightarrow \exists u^*$$

$$\boxed{
\text{An optimal decision exists and is computable.}
}$$

A.7 Closed-Loop Stability

$$\Psi_{k+1} = \Psi_k + dt\,F(\Psi_k, u_k)$$

with:

$$u_k = \pi(\Psi_k)$$

If π is bounded, then F remains Lipschitz.

$$\boxed{\text{The closed-loop system remains stable.}}$$

A.8 Global Closure

$$\mathcal{S}_{k+1} = F_{\mathrm{global}}(\mathcal{S}_k)$$

From previous results:

- Lipschitz continuity
- boundedness

- Lyapunov stability

$$
\Rightarrow
\exists!\ \mathcal{S}_k,\quad
\|\mathcal{S}_k\| \le C
$$

$$
\boxed{
\text{The system is fully closed, stable, and well-posed.}
}
$$

SEKHEM-TSDD Theory — Mohamed Abd Elnaby
DOI: 10.5281/zenodo.19521219